

WHAT IS LERVIT?

LervIT is a mobile-first, two-sided moving marketplace operating in Calgary, Alberta. It connects customers who need to move with vetted freelance movers and enterprise fulfillment partners — matching them by proximity, price, vehicle type, and availability in real time. The platform is designed to scale from individual gig-economy movers to white-label enterprise dispatch contracts.

PLATFORM ARCHITECTURE

Layer	Technology
Frontend	React 18, TypeScript, Vite, Wouter, TanStack Query, Tailwind CSS, shadcn/ui
Backend	Express.js (Node.js, TypeScript, ESM), Zod validation, Pino structured logging
Database	PostgreSQL via Neon (serverless), Drizzle ORM, shared TypeScript schema
Payments	Stripe + Stripe Connect (customer charges, mover escrow payouts, webhook verification)
AI	OpenAI GPT-4o (Vision Engine, Auto-Quote, Support Copilot, Audit Copilot)
Realtime	WebSocket server (mover notifications, token auth, audio alerts)
Comms	Resend (email), Telnyx (SMS + OTP)
Maps	Google Maps Distance Matrix API (proximity matching, ETA, driving distance)
Storage	Object Storage (driver photos, vehicle photos, compliance docs, proof of completion)
Observability	Pino JSON logs, analytics event table, Operations Intelligence dashboard
Background Jobs	node-cron (booking expiry, notification cleanup, abandoned booking reminders)

CORE FEATURES

Customer Side

- Multi-step booking flow with mandatory load photo upload and AI load estimation
- Uber-style proximity matching — top 5 movers ranked by driving distance (15–50 km radius)
- Real-time GPS tracking with live ETA and 6-stage move progress indicators
- Saved payment cards, promo codes (LERVIT20, 20% off first 2 moves), and Stripe-secured checkout
- Abandoned booking recovery with automated email/SMS reminders (up to 3 per booking)

Mover Side

- Job notification system with 10-minute acceptance window and audio alerts
- Stripe Connect onboarding with automated progressive reminders (24 h, 3 days, 7 days)
- Earnings dashboard, payout history, and 7-type driver verification with admin review
- Live GPS sharing (30-second updates, Live badge visible to customers within 1 hour)
- Vehicle classification and smart load-matching (car !' pickup !' van !' truck)

Admin Portal

- Growth dashboard: bookings, revenue, conversion rate, fulfilment hours, 7-day trend chart
- Operations Intelligence: booking funnel, live ops, mover performance leaderboard, revenue cohorts
- Visitor analytics: page views, session tracking, booking funnel from client events
- AI Support Copilot: GPT-4o ticket analysis with priority, root cause, and dual-output responses
- Single-session enforcement for admin accounts

ENTERPRISE PARTNER PORTAL (MOVEDECK V1)

A fully isolated portal at /partner/* for enterprise fulfillment companies (pilot: OOMovers Inc.).

- Role-based access: partner_admin, partner_dispatcher, partner_ops_manager, partner_viewer

- Full booking lifecycle management: accept, reject, assign to driver/crew, status transitions
- Compliance document upload and review workflow (insurance, cargo liability, registration, etc.)
- Incident reporting and resolution tracking
- Team and driver CRUD with vehicle details, photos, and internal notes
- AI Audit Copilot: on-demand GPT-4o analysis per audit entry with insight, recommendation, and risk flag
- Partner-scoped audit log with CSV export and direct admin messaging

Feature	AI CAPABILITIES	Model	Status
Vision Engine 2.0 — furniture ID from photos		GPT-4o Vision	Active
Auto-Quote Predictor — instant price estimate with confidence		GPT-4o	Active
Support Copilot — ticket analysis, priority, dual response		GPT-4o	Active
Audit Copilot — per-entry operational insight + risk flag		GPT-4o	Active
Price Breakdown Explainer		GPT-4o	Feature-flagged off

SECURITY & RELIABILITY

- Stripe webhook signature verification and idempotency checks on all payment events
- Circuit breaker pattern on all external API calls
- Neon pool error listener — FATAL/transient DB errors log a warning and recover without a server restart
- `uncaughtException` handler distinguishes transient infrastructure errors from fatal application errors
- Session-based auth with `connect-pg-simple`, IP-scoped partner data access, production seed endpoint blocked
- All partner routes are scoped to the authenticated partner — cross-tenant data access is not possible

KEY NUMBERS (PRODUCTION — MAY 2026)

Metric	Value
Registered users	71
Total bookings	37
Completed moves	16
Analytics events tracked	913
Enterprise partners onboarded	1 (OOMovers pilot)

LervIT is built and maintained as a monorepo. Frontend, backend, and shared types are fully TypeScript end-to-end. Contact the engineering team for API documentation or integration enquiries.

